

# Amazon S3 Public Access and Block Public Access

## To Begin with the Lab

### Summary of the Lab

In this lab, the complete concept of **Amazon S3 Public Access** was covered. First, it was demonstrated that objects uploaded to an S3 bucket are private by default and cannot be accessed by anonymous users. Next, the two methods of granting public access—**Access Control Lists (ACLs)** and **Bucket Policies**—were introduced. The lab then explained why making a bucket public initially fails because of **Block Public Access** settings. After understanding all four Block Public Access options, the required settings were modified, a Bucket Policy was generated using the AWS Policy Generator, and the bucket was successfully made public. Finally, the public URL was verified using an incognito browser to confirm anonymous access.

### Understanding Amazon S3 Public Access

By default, every object uploaded to an S3 bucket is **private**. Even if someone knows the object URL, AWS returns an **Access Denied** error because only the bucket owner has permission to access it. Public access allows anyone on the internet to read or interact with bucket contents, making it useful for **static website hosting** or **public file sharing**, but it should be used carefully because it exposes data to everyone.

#### Example:

A company wants to share a PDF training guide with employees or customers without requiring AWS credentials. Instead of sending the file manually, the company uploads the PDF to an S3 bucket and grants public read access using a Bucket Policy. Anyone with the object URL can then download the file directly from the internet.

## Two Ways to Grant Public Access

### 1. Access Control Lists (ACLs)

ACLs are permissions attached directly to **individual buckets or objects**. They can make a specific object publicly readable.

#### Important Notes

- ACLs are the older permission model.
- AWS does **not recommend** using ACLs for new implementations.
- By default, ACLs are disabled through **Bucket Owner Enforced** object ownership.
- To use ACLs, Bucket Owner Enforced must first be disabled.

### 2. Bucket Policies (Recommended)

Bucket Policies are **JSON-based access policies** attached to an S3 bucket.

Advantages:

- Centralized permission management
- Easier to maintain
- More flexible than ACLs
- AWS recommended approach

A Bucket Policy defines:

- **Version** – Policy language version
- **Statement** – Collection of policy statements
- **Sid** – Optional identifier
- **Effect** – Allow or Deny
- **Principal** – Who receives permission (\* means everyone)
- **Action** – Allowed S3 action (for example, s3:GetObject)
- **Resource** – Bucket ARN and objects covered by the policy (bucket-name/\*)

## Understanding Block Public Access

AWS protects S3 buckets using **Block Public Access**, which prevents accidental exposure of sensitive data. Even if a valid ACL or Bucket Policy is configured, these settings can override it and stop the bucket from becoming public. AWS recommends leaving Block Public Access enabled unless public access is intentionally required.

## Block Public Access Levels

### 1. Account-Level

- Applies to every bucket in the AWS account.
- Highest priority.
- If enabled, buckets cannot be made public even if bucket-level settings are disabled.

### 2. Bucket-Level

- Applies only to a specific bucket.
- Allows more granular control.
- Can only take effect if Account-Level settings permit it.

### Four Block Public Access Options

#### Option 1 – Block New Public ACLs

- Prevents creation of new public ACLs.
- Existing public ACLs remain unchanged.

#### Option 2 – Block All Public ACLs

- Blocks both existing and future public ACLs.
- Removes the effect of any ACL-based public access.

#### Option 3 – Block New Public Bucket Policies

- Prevents creation of new Bucket Policies that allow public access.
- Existing Bucket Policies continue working.

#### Option 4 – Block All Public Bucket Policies

- Blocks both existing and future Bucket Policies.
- Prevents all public and cross-account access granted through Bucket Policies.
- Provides the strictest level of protection.

## Lab Steps

### Step 1 – Create an S3 Bucket

- Open the Amazon S3 Console.
- Create a new bucket.

**Create bucket** [info](#)

Buckets are containers for data stored in S3.

**General configuration**

**AWS Region**  
US East (N. Virginia) us-east-1

**Bucket type** [info](#)

☒ **General purpose**  
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**  
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

**Bucket namespace**  
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**  
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**  
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

**Bucket name** [info](#)  
cloudfreeks-public-access

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

**Copy settings from existing bucket - optional**  
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

- Keep default settings enabled.
- Upload a sample file into the bucket.
- Copy the object URL.

**cloudfreeks-public-access** [info](#)

**Objects** | Metadata | Properties | Permissions | Metrics | Management | Access Points

**Objects (1/1)**

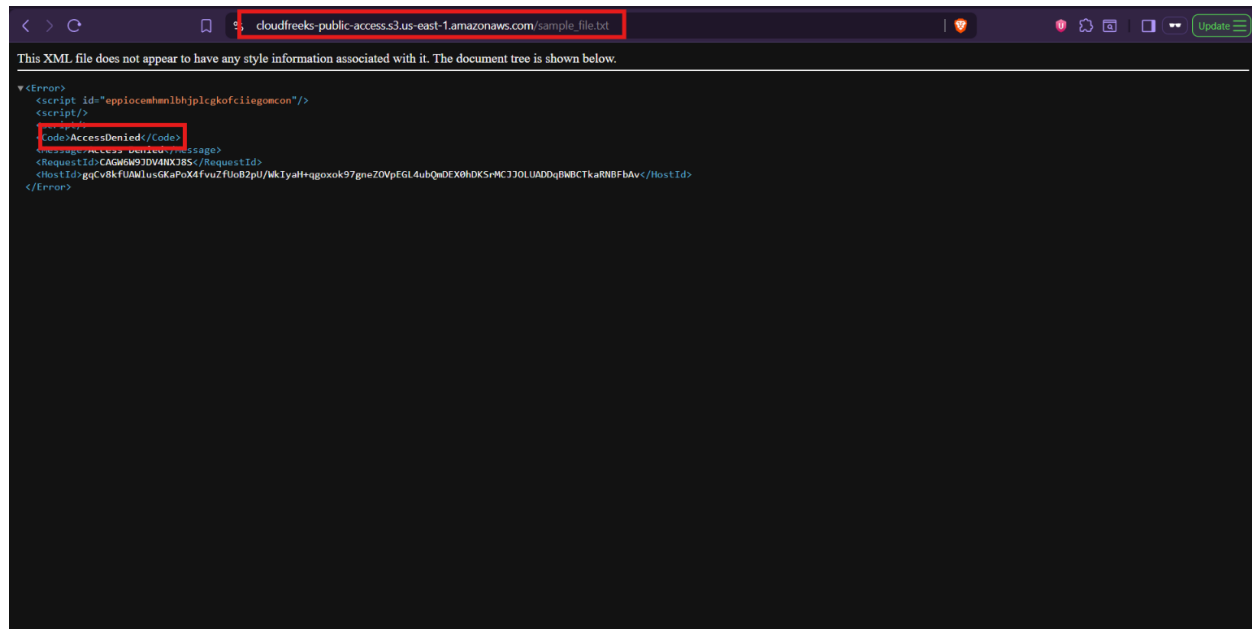
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

| <input checked="" type="checkbox"/> | Name            | Type | Last modified                       | Size   | Storage class |
|-------------------------------------|-----------------|------|-------------------------------------|--------|---------------|
| <input checked="" type="checkbox"/> | sample_file.txt | txt  | July 21, 2026, 18:29:18 (UTC+05:30) | 29.0 B | Standard      |

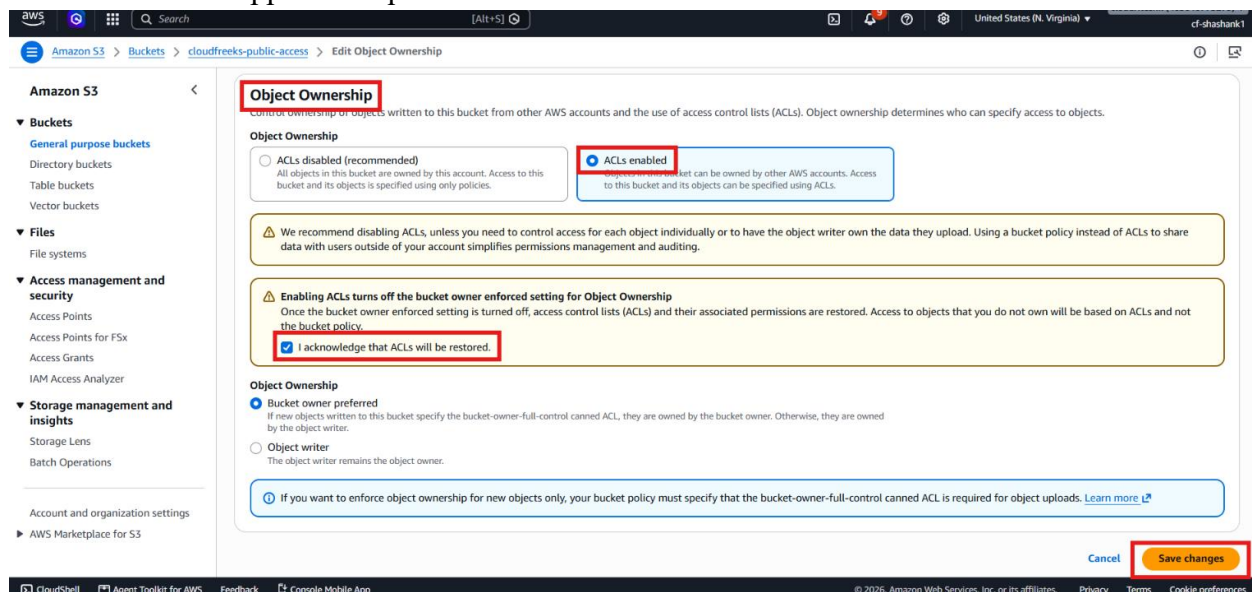
Object URL Copied

- Verify that opening the URL in an incognito browser returns **Access Denied** because the object is private.

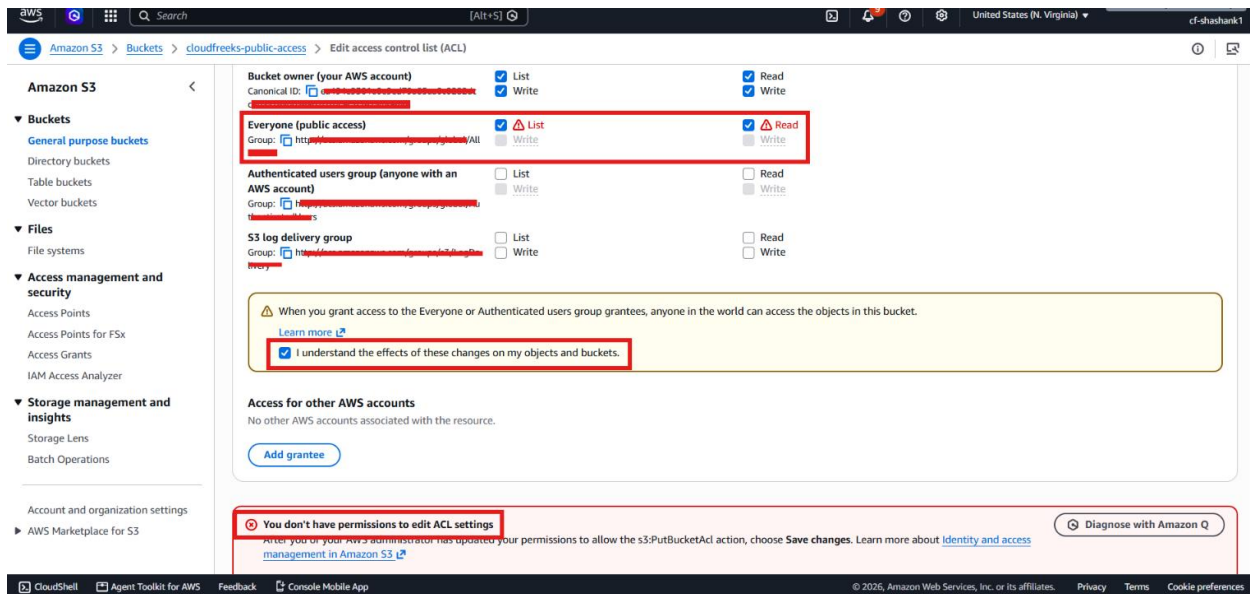


## Step 2 – Explore ACL-Based Public Access

- Navigate to the bucket Permissions tab.
- Observe that ACL editing is disabled because Object Ownership is set to **Bucket Owner Enforced**.
- Enable ACL support if required.

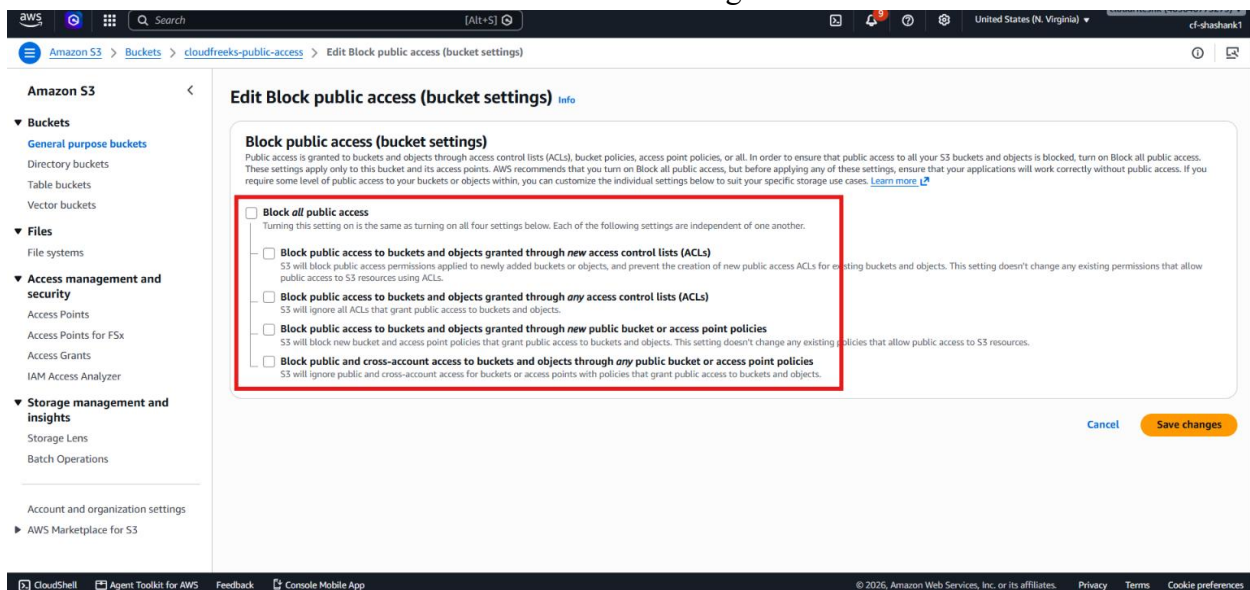


- Attempt to grant public read access.
- Notice that AWS still prevents public access because Block Public Access is enabled.



## Step 3 – Understand Block Public Access

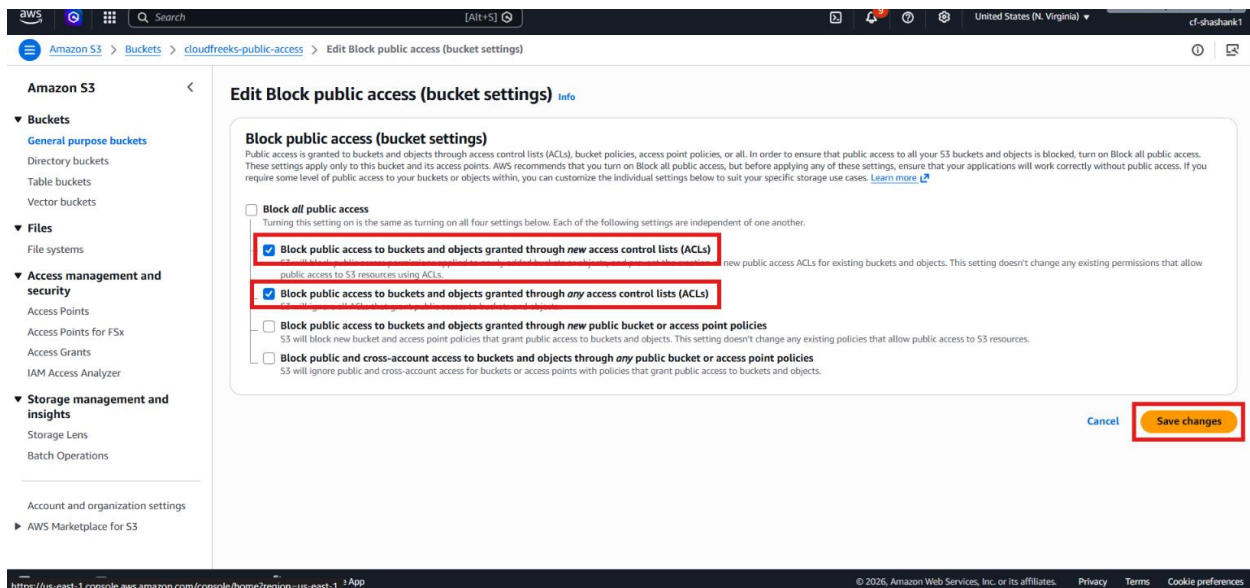
- Open the Block Public Access settings.
- Review both Account-Level and Bucket-Level configurations.



- Understand how AWS prevents accidental exposure of buckets.
- Learn how Account-Level settings override Bucket-Level settings.

## Step 4 – Disable Required Block Public Access Settings

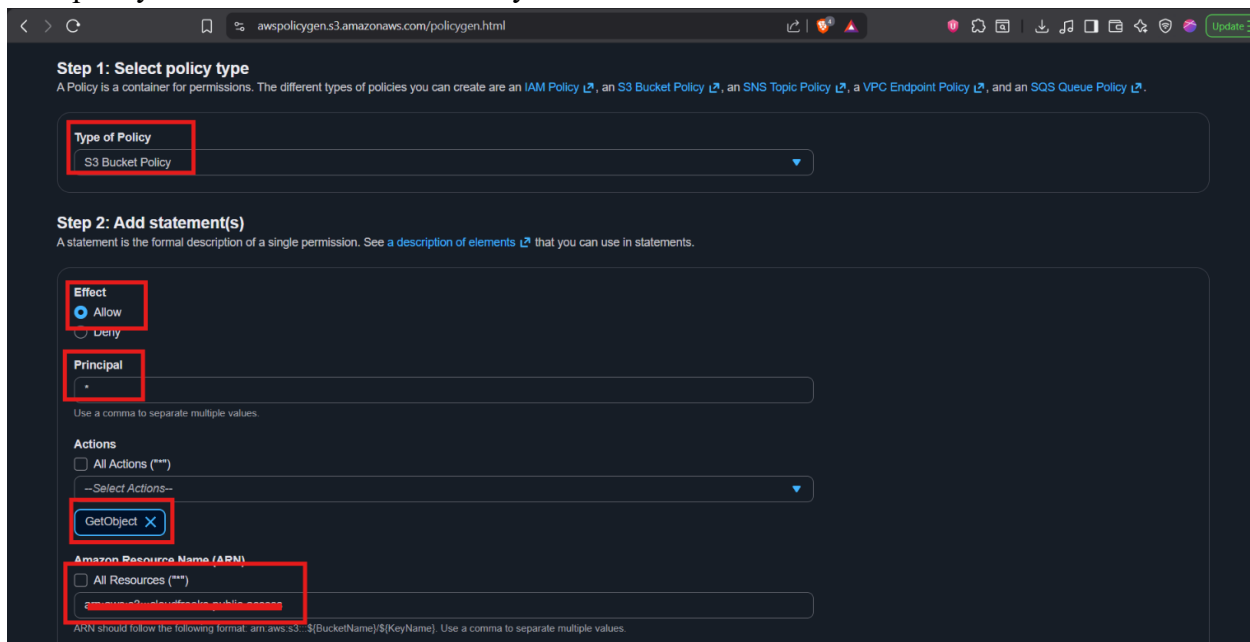
- Open the bucket's Permissions tab.
- Edit Block Public Access settings.
- Disable only the required options that prevent Bucket Policies.
- Save the configuration.



- Confirm the warning message.

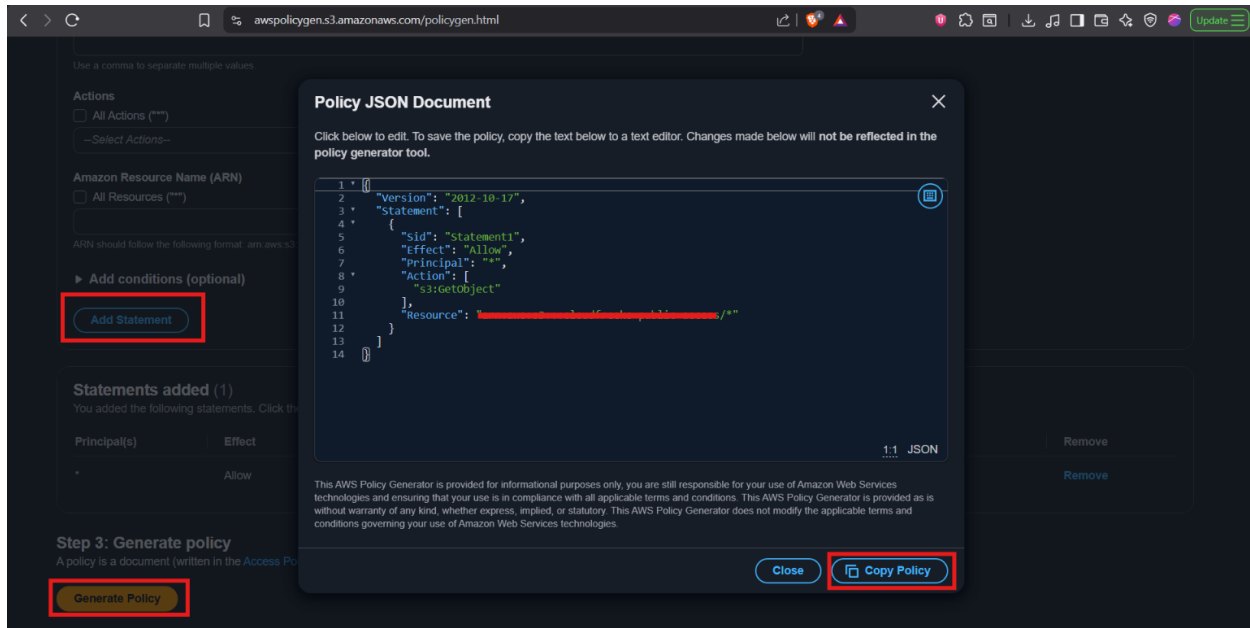
## Step 5 – Generate a Bucket Policy

- Open the AWS Policy Generator.
- <https://awspolicygen.s3.amazonaws.com/policygen.html>
- Select **Amazon S3 Bucket Policy**.
- Choose **Allow**.
- Set Principal to **\***.
- Select the **s3:GetObject** action.
- Specify the bucket ARN followed by **/\***.



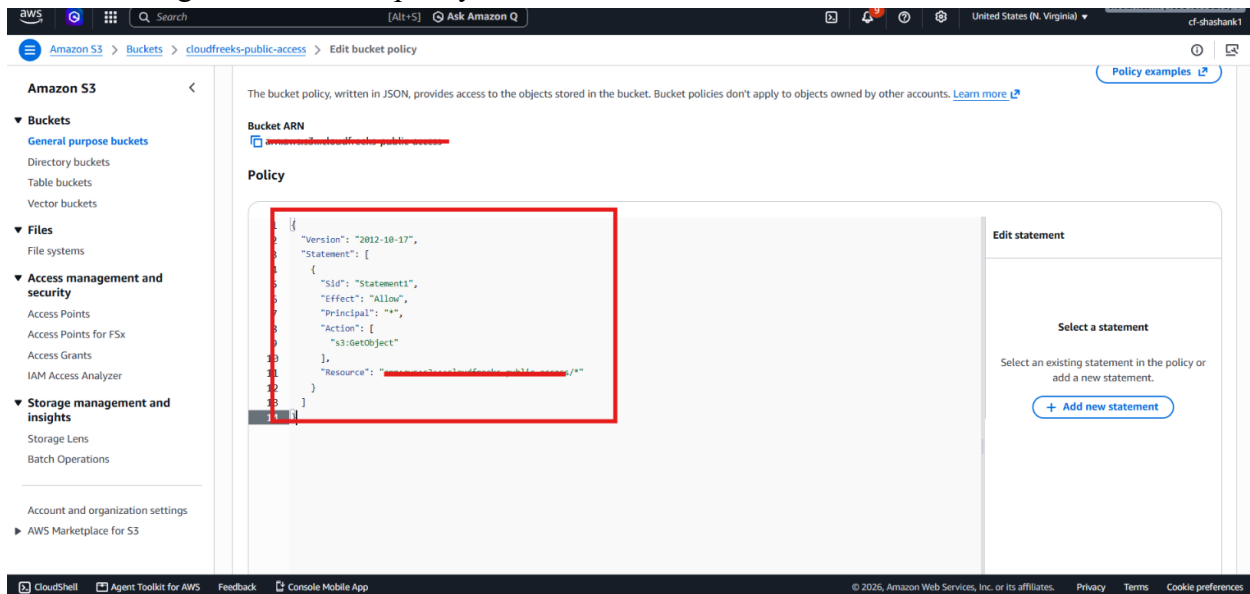
- Generate the JSON Bucket Policy.

- Copy the generated policy.



## Step 6 – Attach the Bucket Policy

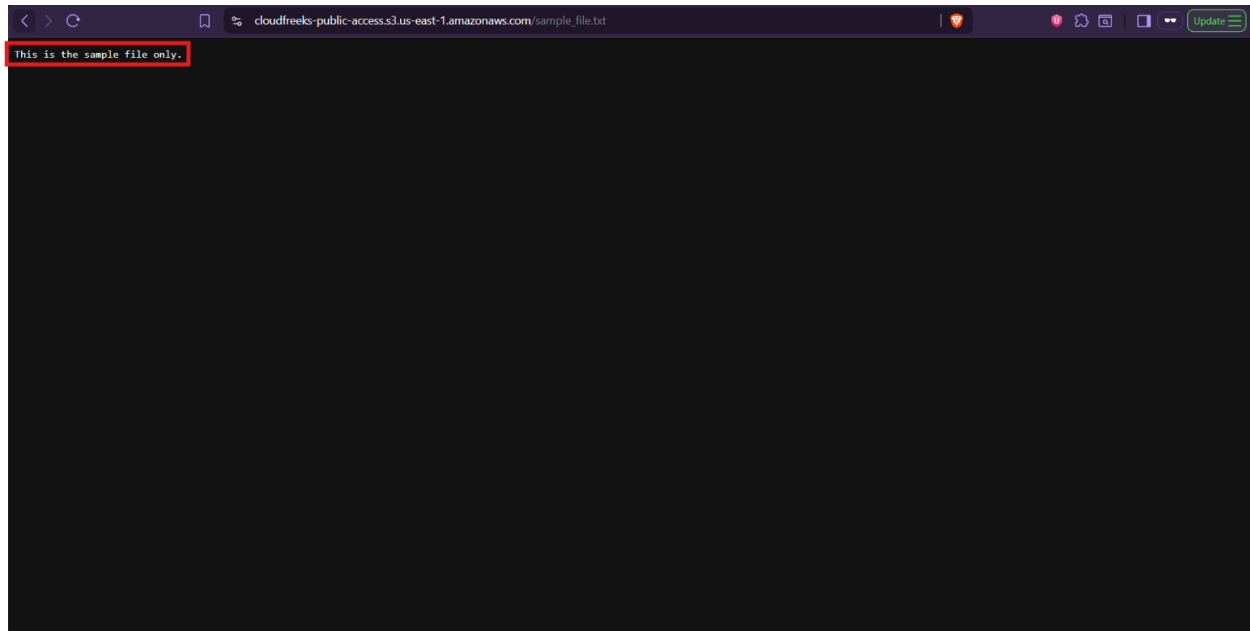
- Return to the S3 bucket.
- Open Bucket Policy.
- Paste the generated JSON policy.



- Save the policy successfully.
- Verify that the bucket now displays **Publicly Accessible**.

## Step 7 – Verify Public Access

- Copy the object URL.
- Open it in an Incognito browser.



- Confirm that the PDF opens successfully without AWS authentication.
- Verify that anyone with the URL can now access the object.

## Verification

- Verified that S3 objects are private by default.
- Verified the difference between ACLs and Bucket Policies.
- Verified that AWS recommends using Bucket Policies instead of ACLs.
- Verified the purpose of Account-Level and Bucket-Level Block Public Access.
- Verified the functionality of all four Block Public Access options.
- Verified that disabling the appropriate Block Public Access settings allows Bucket Policies to grant public access.
- Verified that a Bucket Policy generated using the AWS Policy Generator successfully granted public read access.
- Verified that the uploaded object became publicly accessible through its URL after applying the Bucket Policy.